Information Fusion 127 (2026) 103877

Contents lists available at ScienceDirect

Information Fusion

journal homepage: www.elsevier.com/locate/inffus

Full Length Article

Multi-task mutual learning for multimodal emotion-cause pair extraction in conversations

Geng Tu [a], Jun Wang [a,1], Li Yang [a], Bin Liang [b], Erik Cambria [c], Wenjie Li [d], Ruifeng Xu [a,e,f,*]

[a] School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), Shenzhen postcode, China
[b] Department of Systems Engineering and Engineering Management, The Chinese University of Hong Kong, Hong Kong, China
[c] College of Computing and Data Science, Nanyang Technological University, Singapore
[d] Department of Computing, The Hong Kong Polytechnic University, Hong Kong, China
[e] Peng Cheng Laboratory, Shenzhen postcode, China
[f] Guangdong Provincial Key Laboratory of Novel Security Intelligence Technologies, China

ARTICLE INFO

Keywords:
Multimodal emotion-cause pair extraction in conversations
Multi-task mutual learning
Graph attention network

ABSTRACT

Multimodal Emotion-Cause Pair Extraction (ECPE) in Conversations (MC-ECPE) aims to simultaneously identify emotions and their causes within conversations across different modalities. Early paradigms of ECPE involved a two-step framework for emotion and cause extraction and pairing, resulting in error accumulation. Thus, there is a growing interest in end-to-end ECPE. Despite the progress, emotion, and cause extraction are essentially mutually dependent, yet existing efforts fail to model it deeply. Additionally, Baselines for the MC-ECPE task primarily use traditional fusion methods like concatenation, limiting context understanding between modalities. Based on this, we propose the Multi-Task Mutual Learning (MTML) framework, which utilizes implicit and explicit modeling strategies to model the mutual dependency between emotion and cause extraction. Specifically, we introduce a Multimodal Interactive Graph Attention Network (MIGAT) with three types of connections: intra-modal for conversational context, cross-modal for multimodal fusion, and cross-task for capturing dependencies between emotion and cause extraction tasks. Implicit modeling leverages cross-task connections within MIGAT and information from shared components in progressive multi-task learning (PMTL), while explicit modeling iteratively extracts emotional and causal probability distributions to enhance subsequent reasoning. Experimental results demonstrate the superiority of our MTML over state-of-the-art methods.

1. Introduction

Multimodal Emotion-Cause Pair Extraction (ECPE) in Conversations (MC-ECPE) is first proposed in [1]. MC-ECPE builds upon prior ECPE studies [2] that focused on extracting emotion-cause pairs from textual sources such as news articles [3] or conversations [4], extending its scope to multimodal scenarios. MC-ECPE aims to simultaneously identify emotions and their causes within conversations across different modalities (text, audio, and video).

Early paradigms of ECPE involved a two-step framework [2], wherein the initial step extracts emotions and cause clauses separately, followed by training a binary classifier to identify emotion-cause pairs. Baselines for the MC-ECPE task follow this paradigm. However, this method has two drawbacks: (1) Errors from the first step impact the second step's performance [5] (2) Model training does not directly target the extraction of final emotion-cause pairs [6].

Subsequently, numerous joint models have been proposed employing end-to-end training, broadly falling into three approaches: sliding window-based methods [7,8], pair filtering [5,9–11], and unified labeling [12–15]. The first approach employs a sliding window mechanism to extract emotion-cause pairs. The second approach directly constructs representations of candidate emotion-cause pairs for prediction. In contrast, the third approach transforms the ECPE task into a sequence labeling problem.

Despite the progress, the emotion and cause extraction are essentially mutually dependent [16]. As illustrated in Fig. 1, determining whether the third utterance is a causal clause can be challenging, but knowing that the fourth utterance expresses emotion makes it easier

* Corresponding author.
E-mail addresses: 22b951011@stu.hit.edu.cn (G. Tu), 23s051031@stu.hit.edu.cn (J. Wang), 210710122@stu.hit.edu.cn (L. Yang), bin.liang@cuhk.edu.hk (B. Liang), cambria@ntu.edu.sg (E. Cambria), cswjli@comp.polyu.edu.hk (W. Li), xuruifeng@hit.edu.cn (R. Xu).
[1] The first two authors contribute equally to this work.

https://doi.org/10.1016/j.inffus.2025.103877
Received 1 August 2024; Received in revised form 10 December 2024; Accepted 20 October 2025
Available online 26 October 2025

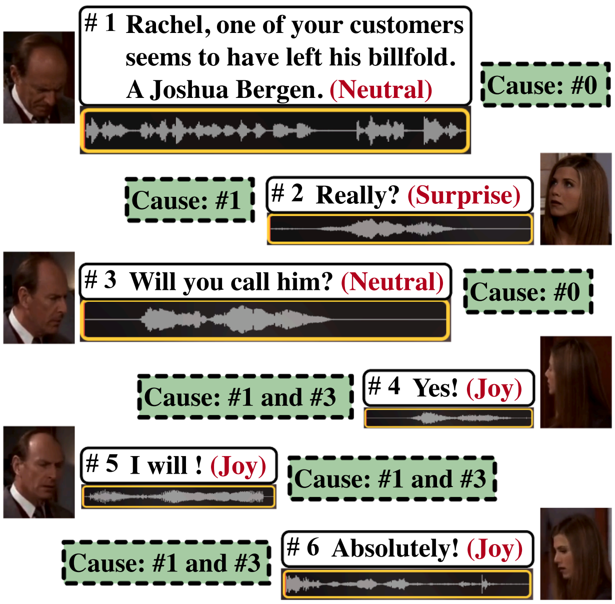

Fig. 1. Examples of utterances for the MC-ECPE task [1]. The emotion of utterances is highlighted in red font, while the corresponding causes are presented in green dashed boxes. #0 denoting neutral emotions without corresponding causes.

to infer the cause from the previous context. While the fourth utterance alone doesn't clearly indicate emotion, causal extraction can aid in emotion extraction. For example, the first utterance mentions someone losing their wallet, and the third asks Rachel if she would call the owner. Thus, the fourth utterance suggests her response may stem from a sense of fulfillment in helping others, reflecting a positive emotion. Existing efforts **fail to model the mutual dependency between them deeply**. Although some end-to-end work in ECPE incorporates shared information within Multi-Task Learning (MTL) frameworks, it fails to explicitly model mutual dependencies [13,17–19]. Instead, it relies on sequentially training one task and then using its outputs to train another, capturing only unidirectional dependencies. For instance, Singh et al. [16] utilize the probability distribution from the emotion extraction task to enhance the cause extraction task.

From a multimodal fusion perspective, current Baselines for the MC-ECPE task **rely solely on traditional fusion methods like concatenation, limiting context understanding between modalities** [20]. Therefore, it is necessary to explore the cross-modal dependencies, which capture relationships between different modalities to extract richer semantic information. Additionally, intra-modal dependencies aid in capturing deeper context features within the same modality. Integrating both types of dependencies enables a more comprehensive understanding of conversations [21].

To address these issues, we propose the Multi-Task Mutual Learning (MTML) framework. It incorporates multi-level Experts with Gate networks to gradually decouple shared and task-specific components at higher levels, ensuring balanced performance across all tasks. To deeply model the mutual dependency between emotion and cause extractions, we employ both implicit and explicit modeling strategies: (1) **Implicit modeling** utilizes shared Experts and cross-task connections within a Multimodal Interactive Graph Attention Network (MIGAT) to extract shared information from the correlated task. MIGAT is designed to model intra-modal and cross-modal dependencies, as well as cross-task dependencies. Cross-modal dependencies capture relationships between different modalities, enriching semantic information, while intra-modal dependencies delve into deeper context features within the same modality. (2) **Explicit modeling** involves a progressive separation routine to extract emotional and causal logits from the lower layer to enhance reasoning at the current layer. Experimental results demonstrate that our method surpasses state-of-the-art methods on the ECF dataset [1]. Our main contributions can be summarized as follows:

- We propose a Multi-modal Interactive Graph Attention Network, which constructs intra-modal connections to model deeper context features within the same modality, cross-modal connections for multimodal fusion, and cross-task connections to leverage shared information from the correlated task.
- To deeply model the mutual dependency between emotion and cause extractions, we integrate implicit modeling through shared experts and cross-task connections in MIGAT with explicit modeling via a progressive separation routine that extracts emotional and causal logits from the lower layer.
- Experimental findings on the ECF dataset indicate that our method outperforms state-of-the-art methods.

2. Related work

Emotion Cause Analysis: Emotion cause extraction (ECE) was first introduced by Lee et al. [22], to identify cause spans associated with a given emotion within a text. Following this, some researchers have employed machine learning techniques [23,24] or rule-based methods [25–28] to extract emotion causes in their corpus. Analyzing the corpus from Lee et al. [26], Chen et al. [29] suggested that clauses might serve as a more appropriate unit for annotating causes of emotions, suggesting that extracting emotion causes at the clause level might be more effective. Subsequent research has adhered to this task setting [30,31]. Notably, Gui et al. [3] released an open Chinese emotion cause dataset, which has become the benchmark for the ECE task. This dataset has spurred extensive research, leading to the development of both traditional machine learning methods[3,32] and deep learning approaches [33–37].

Emotion-Cause Pair Extraction: The ECE task faces two significant challenges: (1) emotions must be manually annotated before extracting causes, limiting its practical application, and (2) annotating emotions independently before extracting causes fails to account for the interdependence between emotions and causes. To tackle these issues, Xia and Ding [2] introduced the ECPE task, which has garnered considerable attention [5,14,15,19,38]. Xia and Ding [2] introduced a pipelined approach for the ECPE task, first extracting emotions and causes separately, then pairing and filtering them. However, this method has two drawbacks: (1) errors in the first step can affect the second step's performance [5] and (2) model training does not directly focus on extracting final emotion-cause pairs [6].

In response, several joint models employing end-to-end training have been developed, generally falling into three categories: the sliding window-based approach [7,8], the pair filtering approach [5,9–11,18,39,40], and the unified labeling approach [12–15,41,42]. The first approach employs a sliding window mechanism to extract emotion-cause pairs. Cheng et al.[7] explored emotions or causes related to the central clauses within a sliding window framework, while Ding et al.[8] posited that each clause could represent either an emotion or a cause, identifying corresponding pairs within the sliding window. The second approach directly constructs representations of candidate emotion-cause pairs for prediction. For example, Ding et al.[9] proposed a unified framework that incorporates emotion-cause pair representation, interaction, and prediction. In contrast, Wu et al.[18] introduced a multi-task approach for extracting emotion-cause pairs. The third approach transforms the ECPE task into a sequence labeling problem. Yuan et al. [12] introduced a method for encoding the relative distances between emotions and their causes into unified labels for pairing. Building on this, Fan et al. [10] further enhanced these labels by incorporating the results from individual extraction of emotions and causes.

Despite the progress, extracting emotions and their causes remains fundamentally mutually dependent. Current approaches fail to adequately model this mutual dependency at a deep level, which may lead to misjudgments by the model in the MC-ECPE task.

3. Methodology

In this section, we provide a detailed introduction to each component of the MTML, as depicted in Fig. 2.

3.1. Task definition

Let a conversation $\mathbf{D} = [\mathbf{U}_1, \ldots, \mathbf{U}_i, \ldots, \mathbf{U}_\mathbf{N}]$ consisting of $\mathbf{N}$ utterances. Each utterance $\mathbf{U}_i$ is represented by a triplet $\{\mathbf{x}_i^a; \mathbf{x}_i^v; \mathbf{x}_i^t\}$. $\mathbf{x}_i^a \in \mathbb{R}^{d_a}$, $\mathbf{x}_i^v \in \mathbb{R}^{d_v}$, and $\mathbf{x}_i^t \in \mathbb{R}^{d_t}$ denote the acoustic, visual, and text features of $\mathbf{U}_i$, respectively. MC-ECPE aims to extract a set of emotion-cause pairs $\mathbf{P}$ in a conversational video.

$$\mathbf{P} = \{\ldots, (\mathbf{U}_j^e - \mathbf{U}_k^c), \ldots\} \tag{1}$$

In the emotion-cause pair $(\mathbf{U}_j^e - \mathbf{U}_k^c)$, where $\mathbf{U}_j^e$ represents an emotion utterance and $\mathbf{U}_k^c$ denotes the corresponding cause utterance.

3.2. Feature representation

Following [1], we employ a pre-trained BERT model [43] to encode the text, processing each utterance independently to derive its textual features, $\mathbf{x}_i^t$. For audio, we utilize the 6373-dimensional acoustic features, $\mathbf{x}_i^a$, derived by Poria et al. [44] using openSMILE [45] within the MELD dataset [44]. For video, we employ a 3D-CNN network, namely C3D [46], to extract 128-dimensional visual features, denoted as $\mathbf{x}_i^v$. In detail, we extract 16 frames, each with a resolution of 171×128, and input these frames into the C3D network to obtain a 4096-dimensional video descriptor.

3.3. Utterance-level encoder

To handle the inconsistent dimensions in multimodal data, we use a fully connected layer $\mathbf{FCL}^\eta \in \mathbb{R}^{d_h \times d_{a/v/t}}$ for acoustic, visual, and textual modalities, to map the feature sequence $\mathbf{x}_i^\eta \in \mathbb{R}^{d_{a/v/t}}$ of each modality $\eta \in \{a, v, t\}$ to a fixed-size representation $\mathbf{m}_i^\eta \in \mathbb{R}^{d_h}$.

$$\mathbf{m}_i^\eta = \mathbf{FCL}^\eta(\mathbf{x}_i^\eta \mid \theta^\eta) \tag{2}$$

where $\mathbf{FCL}^\eta$ assigns separate parameters θ^η for each modality. For the MIGAT, $\mathbf{m}_i^\eta$ is the input, denoted as $\mathbb{H}_i^{\eta,\psi,0}$, where $\psi \in \{e, c\}$ represents the emotion and cause extraction.

3.4. Multi-modal interactive GAT

In this section, we construct two sets of three conversation subgraphs for acoustic, visual, and text modalities, focusing on emotion and cause extraction tasks. MIGAT, comprising $\mathbf{N}_\xi$ layers, integrates conversational context, multi-modal fusion, and multi-task interaction through intra-modal, cross-modal, and cross-task connections.

Graph Structure: A conversation with $\mathbf{N}$ utterances can be represented as a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V}$ ($|\mathcal{V}| = 6\mathbf{N}$) denotes the utterance nodes in emotion and cause extraction tasks, across three modalities. $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ is a set of relationships: context, modality and task dependencies.

Nodes: ith utterance in the ξth layer can be represented by a triplet for the emotion extraction task: $\{\mathbb{H}_i^{a,e,\xi}, \mathbb{H}_i^{v,e,\xi}, \mathbb{H}_i^{t,e,\xi}\}$, and for the cause extraction task: $\{\mathbb{H}_i^{a,c,\xi}, \mathbb{H}_i^{v,c,\xi}, \mathbb{H}_i^{t,c,\xi}\}$.

Edges: Each node $\mathbb{H}_i^{\eta,\psi,\xi}$ has three types of edges. **(1) Intra-modal connections:** We contend that each utterance is contextually dependent on its neighboring utterances, including itself. To balance computational cost and efficiency, we designate a specific window size ω for both the past and future context, rather than connecting it to all other utterances. Each node $\mathbb{H}_i^{\eta,\psi,\xi}$ has its nearest neighbor node $\mathbb{H}_k^{\eta,\psi,\xi}$ in the same modality, where $|k - i| \le \omega$. $(\mathbb{H}_i^{\eta,\psi,\xi}, \mathbb{H}_k^{\eta,\psi,\xi}, \boldsymbol{\alpha}_{i,k}^{\eta,\psi,\xi}) \in \mathcal{E}$ denotes the intra-modal connections with weight $\boldsymbol{\alpha}_{i,k}^{\eta,\psi,\xi}$. **(2) Cross-modal connections:** To learn multi-modal complementarity, 3 nodes of the utterance i are interconnected across different modalities. $(\mathbb{H}_i^{\eta_1,\psi,\xi}, \mathbb{H}_i^{\eta_2,\psi,\xi}, \boldsymbol{\alpha}_{i,i}^{\eta_2,\psi,\xi}) \in \mathcal{E}$ denotes the cross-modal connections with weight $\boldsymbol{\alpha}_{i,i}^{\eta_2,\psi,\xi}$, where $\eta_1 \ne \eta_2$. Moreover, there are also connections between $\mathbb{H}_i^{\eta_1,\psi,\xi}$ and $\mathbb{H}_k^{\eta_2,\psi,\xi}$. **(3) Cross-task connections:** Due to the close relationship between emotion extraction and cause extraction in MC-ECPE, these connections link nodes between $\mathbb{H}_i^{\eta,\psi_1,\xi}$ and $\mathbb{H}_i^{\eta,\psi_2,\xi}$, as well as their neighboring nodes $\mathbb{H}_k^{\eta,\psi_2,\xi}$ within modality η to extract shared information from the correlated task. For nodes $\mathbb{H}_i^{\eta,\psi_1,\xi}$ and $\mathbb{H}_k^{\eta,\psi_2,\xi}$, these connections include edges $(\mathbb{H}_i^{\eta,\psi_1,\xi}, \mathbb{H}_k^{\eta,\psi_2,\xi}, \boldsymbol{\alpha}_{i,k}^{\eta,\psi_2,\xi}) \in \mathcal{E}$, where $\psi_1 \ne \psi_2$.

Update: The update mechanism of utterance representation via GAT is illustrated using node $\mathbb{H}_i^{t,e,\xi}$ in the emotion extraction task as an example, which is similar across other modalities and the cause extraction task. $\mathbb{H}_i^{t,e,\xi}$ is obtained by aggregating messages $\widehat{\mathbb{H}}_{i,k}^{t,e,\xi-1}$ from intra-modal nodes, messages $\widehat{\mathbb{H}}_{i,k}^{v,e,\xi-1}$ and $\widehat{\mathbb{H}}_{i,k}^{a,e,\xi-1}$ from cross-modal nodes, and messages $\widehat{\mathbb{H}}_{i,k}^{t,c,\xi-1}$ from cross-task nodes.

$$\mathbb{H}_i^{t,e,\xi} = \sum_{|k-i| \le \omega} (\widehat{\mathbb{H}}_{i,k}^{t,e,\xi-1} + \widehat{\mathbb{H}}_{i,k}^{v,e,\xi-1} + \widehat{\mathbb{H}}_{i,k}^{a,e,\xi-1} + \widehat{\mathbb{H}}_{i,k}^{t,c,\xi-1}) \tag{3}$$

$$\widehat{\mathbb{H}}_{i,k}^{\eta,\psi,\xi-1} = \alpha_{i,k}^{\eta,\psi,\xi-1} \mathbf{W}_h^{\eta,\psi,\xi-1} \mathbb{H}_k^{\eta,\psi,\xi-1} \tag{4}$$

where, $\mathbf{W}_h^{\eta,\psi,\xi-1} \in \mathbb{R}^{d_h \times d_h}$ represent projection parameters. $\alpha_{i,k}^{\eta,\psi,\xi-1}$ represents the attention weight for the message aggregation mechanism in GAT. It quantifies the importance of the message from node $\mathbb{H}_k^{\eta,\psi,\xi-1}$ to node $\mathbb{H}_i^{t,e,\xi-1}$, based on the correlation of their representations. This attention weight is computed using a softmax function applied over the correlation score $s_{i,k}^{\eta,\psi,\xi-1}$ between nodes $\mathbb{H}_i^{t,e,\xi-1}$ and $\mathbb{H}_k^{\eta,\psi,\xi-1}$.

$$\alpha_{i,k}^{\eta,\psi,\xi-1} = \frac{\exp(\mathbf{LeakyReLU}(s_{i,k}^{\eta,\psi,\xi-1}))}{\sum_{|j-i| \le \omega} \exp(\mathbf{LeakyReLU}(s_{i,j}^{\eta,\psi,\xi-1}))} \tag{5}$$

$$s_{i,k}^{\eta,\psi,\xi-1} = (\mathbf{a}_I^{\eta,\psi,\xi-1})^\top ([\mathbf{W}_I^{\eta,\psi,\xi-1} \mathbb{H}_i^{t,e,\xi-1}; \mathbf{W}_I^{\eta,\psi,\xi-1} \mathbb{H}_k^{\eta,\psi,\xi-1}]) \tag{6}$$

where $[;]$ denotes the concatenation operation. $\mathbf{W}_I^{\eta,\psi,\xi-1} \in \mathbb{R}^{d_h \times d_h}, \mathbf{a}_I^{\eta,\psi,\xi-1} \in \mathbb{R}^{2d_h}$ are trainable parameters used to compute the correlation. It can be observed that nodes connected by different types of edges use different parameters to compute the correlation.

Output: We concatenate the representations of three modalities from the final layer of MIGAT to obtain the multimodal features $\mathbb{M}_i^e, \mathbb{M}_i^c \in \mathbb{R}^{3d_h}$ for emotion extraction and cause extraction, respectively.

$$\mathbb{M}_i^e = [\mathbb{H}_i^{t,e,\mathbf{N}_\xi}; \mathbb{H}_i^{v,e,\mathbf{N}_\xi}; \mathbb{H}_i^{a,e,\mathbf{N}_\xi}] \tag{7}$$

$$\mathbb{M}_i^c = [\mathbb{H}_i^{t,c,\mathbf{N}_\xi}; \mathbb{H}_i^{v,c,\mathbf{N}_\xi}; \mathbb{H}_i^{a,c,\mathbf{N}_\xi}] \tag{8}$$

where $[;]$ denotes the concatenation operation.

3.5. Progressive multi-task learning

The progressive multi-task learning (PMTL) framework consists of 3 Experts and 3 Gate networks for extracting emotions, causes, and emotion-cause pairs at each layer. This structure effectively distinguishes between shared and task-specific experts to reduce harmful parameter interference between them.

Input Layer: Project $\mathbb{M}_i^e$, $\mathbb{M}_i^c$ and $[\mathbb{M}_i^e; \mathbb{M}_i^c]$ using linear layers to produce inputs $\mathcal{H}_i^{e,(0)}$, $\mathcal{H}_i^{c,(0)}$ and $\mathcal{H}_i^{s,(0)}$ for the emotion Expert $\mathbf{Expert}_e$, cause Expert $\mathbf{Expert}_c$ and shared Expert $\mathbf{Expert}_s$.

$$\mathcal{H}_i^{e,(0)} = \mathbb{M}_i^e \mathbf{W}_e + \mathbf{b}_e \tag{9}$$

$$\mathcal{H}_i^{c,(0)} = \mathbb{M}_i^c \mathbf{W}_c + \mathbf{b}_c \tag{10}$$

$$\mathcal{H}_i^{s,(0)} = [\mathbb{M}_i^e; \mathbb{M}_i^c] \mathbf{W}_s + \mathbf{b}_s \tag{11}$$

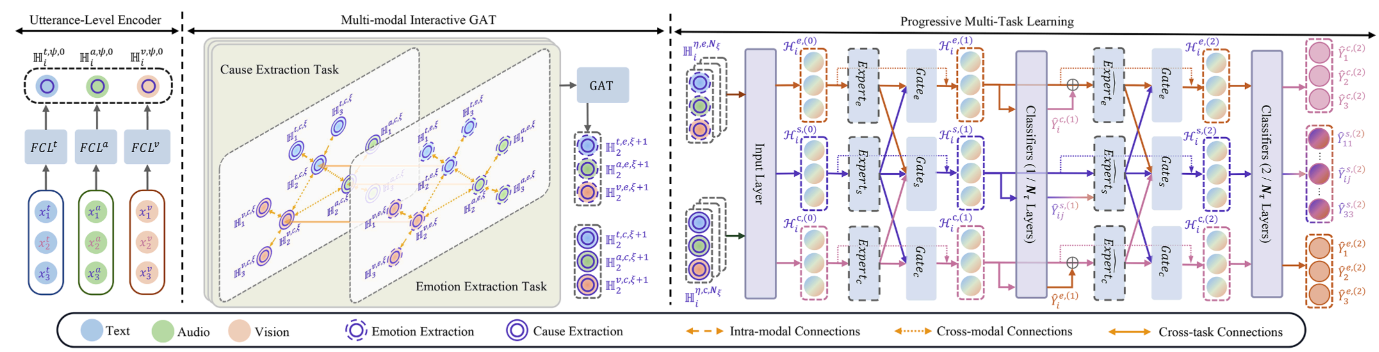

Fig. 2. The proposed MTML framework, exemplified by the update to utterance $\mathbf{U}_2$ in MIGAT, showcases Progressive MTL from layers 0 to 1. Mathematical symbols are consistent with the formulas in the paper. To model the mutual dependency between emotion and cause extractions, we use both implicit and explicit strategies: (1) **Implicit modeling** utilizes shared Experts and cross-task connections within a MIGAT to extract shared information. (2) **Explicit modeling** involves a progressive separation routine to extract emotional and causal probability distributions from the lower layer to enhance reasoning at the current layer.

where $\mathbf{W}_e \in \mathbb{R}^{3d_h\times d_h}$, $\mathbf{W}_c \in \mathbb{R}^{3d_h\times d_h}$, and $\mathbf{W}_s \in \mathbb{R}^{6d_h\times d_h}$ represent projection parameters. $\mathbf{b}_e$, $\mathbf{b}_c$, and $\mathbf{b}_s$ are biases.

Expert: The $\mathbf{Expert}_\Psi$ is a self-attention network [47] enhanced with Rotational Positional Encoding (RoPE) [48]. In the τth layer of the PMTL framework, $\mathbf{Expert}_\Psi$ transforms input $\mathcal{H}_i^{\Psi,(\tau-1)}$ into output $\mathcal{O}_i^{\Psi,(\tau)} \in \mathbb{R}^{d_h}$, where $\Psi \in \{e,c,s\}$. In the first layer ($\tau = 1$), the computation process is as follows.

$$\mathcal{O}_i^{\Psi,(\tau)} = \sum_{k=1}^{\mathbf{N}} \mathcal{A}_{ik}^{\Psi,(\tau)} \mathcal{V}_k^{\Psi,(\tau)} + \mathcal{H}_i^{\Psi,(\tau-1)} \tag{12}$$

$$\mathcal{A}_{ik}^{\Psi,(\tau)} = \frac{\exp(\mathcal{Q}_i^{\Psi,(\tau)}(\mathcal{K}_k^{\Psi,(\tau)})^\top)}{\sum_{j=1}^{\mathbf{N}} \exp(\mathcal{Q}_i^{\Psi,(\tau)}(\mathcal{K}_j^{\Psi,(\tau)})^\top)} \tag{13}$$

$$\mathcal{Q}_i^{\Psi,(\tau)} = \mathcal{H}_i^{\Psi,(\tau-1)} \mathbf{W}_Q^{\Psi,(\tau)} \mathcal{R}_i \tag{14}$$

$$\mathcal{K}_i^{\Psi,(\tau)} = \mathcal{H}_i^{\Psi,(\tau-1)} \mathbf{W}_K^{\Psi,(\tau)} \mathcal{R}_i \tag{15}$$

$$\mathcal{V}_i^{\Psi,(\tau)} = \mathcal{H}_i^{\Psi,(\tau-1)} \mathbf{W}_V^{\Psi,(\tau)} \tag{16}$$

where $\mathbf{W}_Q^{\Psi,(\tau)}, \mathbf{W}_K^{\Psi,(\tau)}, \mathbf{W}_V^{\Psi,(\tau)} \in \mathbb{R}^{d_h\times d_h}$ are the projection parameters. $\mathcal{R}_i \in \mathbb{R}^{d_h\times d_h}$ is the rotation matrix [48].

In layers beyond the first ($\tau \geq 2$), $\widehat{\mathbf{Expert}}_\psi$ differs from the first layer only in its input, which extracts emotional and causal probability distributions from the lower layer, serving as a form of knowledge to enhance reasoning at the current layer, where $\psi \in \{e,c\}$. This change occurs from $\mathcal{H}_i^{e,(\tau-1)}$ to $[\mathcal{H}_i^{e,(\tau-1)}; \mathbf{Y}_i^{c,(\tau-1)}]$ or $\mathcal{H}_i^{c,(\tau-1)}$ to $[\mathcal{H}_i^{c,(\tau-1)}; \mathbf{Y}_i^{e,(\tau-1)}]$. $\widehat{\mathbf{Expert}}_s$ is a modified version of $\mathbf{Expert}_s$, employing an additional head through external score $\mathcal{B}^{(\tau-1)} \in \mathbb{R}^{\mathbf{N}\times\mathbf{N}}$, enhancing its capabilities.

$$\mathcal{O}_i^{s,(\tau)} = \sum_{k=1}^{\mathbf{N}} [\mathcal{A}_{ik}^{s,(\tau)} \mathcal{V}_{1k}^{s,(\tau)}; \mathcal{B}_{ik}^{(\tau-1)} \mathcal{V}_{2k}^{s,(\tau)}] + \mathcal{H}_i^{s,(\tau-1)} \tag{17}$$

$$\mathcal{V}_{1k}^{s,(\tau)} = \mathcal{H}_k^{s,(\tau-1)} \mathbf{W}_{V_1}^{s,(\tau)} \tag{18}$$

$$\mathcal{V}_{2k}^{s,(\tau)} = \mathcal{H}_k^{s,(\tau-1)} \mathbf{W}_{V_2}^{s,(\tau)} \tag{19}$$

where $\mathbf{W}_{V_1}^{s,(\tau)}, \mathbf{W}_{V_2}^{s,(\tau)} \in \mathbb{R}^{d_h\times(d_h/2)}$ are projection parameters in two attention heads, respectively.

$\mathcal{B}^{(\tau-1)}$ is from the previous layer's prediction.

$$\mathcal{B}_{ik}{}^{(\tau-1)} = \frac{\exp(\boldsymbol{\beta}_{i,k}^{(\tau-1)})}{\sum_{j=1}^{\mathbf{N}} \exp(\boldsymbol{\beta}_{i,j}^{(\tau-1)})} \tag{20}$$

$$\boldsymbol{\beta}_{i,j}^{(\tau-1)} = \hat{\mathbf{Y}}_{i,j}^{s,(\tau-1)}[1] - \hat{\mathbf{Y}}_{i,j}^{s,(\tau-1)}[0] \tag{21}$$

where $\hat{\mathbf{Y}}^{s,(\tau-1)} \in \mathbb{R}^{\mathbf{N}\times\mathbf{N}\times 2}$ is the logits for extracting emotion-cause pairs. $\hat{\mathbf{Y}}_{i,j}^{s,(\tau-1)}[1]$ and $\hat{\mathbf{Y}}_{i,j}^{s,(\tau-1)}[0]$ from layer $\tau-1$ represent the confidence for predicting whether $\mathbf{U}_i - \mathbf{U}_j$ is a true pair or not.

Gate Network: The network aggregates input from various Experts and employs attention with a trainable task-specific vector $\mathcal{S}^{\Psi,(\tau)} \in \mathbb{R}^{d_h}$, randomly initialized, to produce the output $\widehat{\mathcal{O}}_i^{\Psi,(\tau)} \in \mathbb{R}^{d_h}$.

$$\boldsymbol{\Pi}_i^{K,e,(\tau)} = [\mathbf{W}_e^{K,(\tau)} \mathcal{O}_i^{e,(\tau)}, \mathbf{W}_s^{K,(\tau)} \mathcal{O}_i^{s,(\tau)}] \tag{22}$$

$$\boldsymbol{\Pi}_i^{V,e,(\tau)} = [\mathbf{W}_e^{V,(\tau)} \mathcal{O}_i^{e,(\tau)}, \mathbf{W}_s^{V,(\tau)} \mathcal{O}_i^{s,(\tau)}] \tag{23}$$

$$\widehat{\mathcal{O}}_i^{e,(\tau)} = \mathbf{Softmax}\left(\mathcal{S}^{e,(\tau)}(\boldsymbol{\Pi}_i^{K,e,(\tau)})^\top\right)\boldsymbol{\Pi}_i^{V,e,(\tau)} \tag{24}$$

where $\mathbf{W}_e^{K,(\tau)}, \mathbf{W}_s^{K,(\tau)}, \mathbf{W}_e^{V,(\tau)}, \mathbf{W}_s^{V,(\tau)} \in \mathbb{R}^{d_h\times d_h}$ are projection parameters. $\widehat{\mathcal{O}}_i^{c,(\tau)}$ is derived from $\mathcal{O}_i^{c,(\tau)}$ and $\mathcal{O}_i^{s,(\tau)}$ in a similar manner. For $\widehat{\mathcal{O}}_i^{s,(\tau)}$, the acquisition process is slightly different as it requires integrating outputs $\mathcal{O}_i^{e,(\tau)}$, $\mathcal{O}_i^{c,(\tau)}$, and $\mathcal{O}_i^{s,(\tau)}$ from all Experts.

3.6. Model training

The probability distributions for emotion and cause extraction in the ith utterance are represented as $\hat{\mathbf{Y}}_i^{e,(\tau)}$ and $\hat{\mathbf{Y}}_i^{c,(\tau)}$, respectively.

$$\mathcal{H}_i^{\psi,(\tau)} = \widehat{\mathcal{O}}_i^{\psi,(\tau)} + \mathcal{H}_i^{\psi,(\tau-1)} \tag{25}$$

$$\hat{\mathbf{Y}}_i^{\psi,(\tau)} = \mathbf{Softmax}(\mathcal{H}_i^{\psi,(\tau)} \mathbf{W}^{\psi,(\tau)} + \mathbf{b}^{\psi,(\tau)}) \tag{26}$$

where $\psi \in \{e,c\}$. $\mathbf{W}^{\psi,(\tau)} \in \mathbb{R}^{d_h\times 2}$ represents projection parameters. $\mathbf{b}^{\psi,(\tau)}$ is the bias. In the pair extraction, for $\mathbf{U}_i$ as the emotion utterance and $\mathbf{U}_j$ as the cause utterance, the feature representation $\mathbf{X}_{i,j}^{pair,(\tau)} \in \mathbb{R}^{4\times d_h+d_p}$ is constructed by concatenating the corresponding feature vectors and the relative position embedding.

$$\mathcal{H}_i^{s,(\tau)} = \widehat{\mathcal{O}}_i^{s,(\tau)} + \mathcal{H}_i^{s,(\tau-1)} \tag{27}$$

$$\mathcal{H}_{i,j}^{pair,(\tau)} = [\mathcal{H}_i^{e,(\tau)}; \mathcal{H}_i^{s,(\tau)}; \mathcal{H}_j^{c,(\tau)}; \mathcal{H}_j^{s,(\tau)}; \mathbf{rpe}_{i,j}] \tag{28}$$

where $\mathbf{rpe}_{i,j} \in \mathbb{R}^{d_p}$ represents the relative position embedding of the ith utterance relative to the jth utterance.

$$\hat{\mathbf{Y}}_{i,j}^{s,(\tau)} = \mathcal{H}_{i,j}^{pair,(\tau)} \mathbf{W}^{pair,(\tau)} + \mathbf{b}^{pair,(\tau)} \tag{29}$$

$$\hat{\mathbf{Y}}_{i,j}^{pair,(\tau)} = \mathbf{Softmax}(\hat{\mathbf{Y}}_{i,j}^{s,(\tau)}) \tag{30}$$

where $\mathbf{W}^{pair,(\tau)} \in \mathbb{R}^{(4\times d_h+d_p)\times 2}$ is the projection parameter. $\mathbf{b}^{pair,(\tau)}$ is the bias. $\hat{\mathbf{Y}}_{i,j}^{pair,(\tau)} \in \mathbb{R}^2$ is the predicted probability of the pair $\mathbf{U}_i - \mathbf{U}_j$ being the emotion-cause pair. It's worth noting that we utilize the results from the final layer (the $\mathbf{N}_\tau$th layer) as the ultimate prediction outcome.

Loss items: Our MTML is trained through a joint training process, minimizing three loss items at each layer.

$$\mathcal{L}_{all}^{(\tau)} = \mathcal{L}_{emo}^{(\tau)} + \mathcal{L}_{cau}^{(\tau)} + \mathcal{L}_{pair}^{(\tau)} \tag{31}$$

$$\mathcal{L}_{emo}^{(\tau)} = -\sum_{i=1}^{\mathbf{N}} \mathbf{Y}_i^e \cdot \mathbf{log}(\hat{\mathbf{Y}}_i^{e,(\tau)}) \tag{32}$$

$$\mathcal{L}_{cau}^{(\tau)} = -\sum_{i=1}^{\mathbf{N}} \mathbf{Y}_i^c \cdot \mathbf{log}(\hat{\mathbf{Y}}_i^{c,(\tau)}) \tag{33}$$

$$\mathcal{L}^{(\tau)}_{pair} = -\sum\nolimits_{i=1}^{\mathbf{N}}\sum\nolimits_{j=1}^{\mathbf{N}} \mathbf{Y}^{pair}_{i,j}\cdot \mathbf{log}(\widehat{\mathbf{Y}}^{pair,(\tau)}_{i,j}) \tag{34}$$

where $\mathcal{L}_{emo}$, $\mathcal{L}_{cau}$, and $\mathcal{L}_{pair}$ denote the classification loss for the emotion extraction, the cause extraction, and the emotion-cause pair extraction, respectively. The total loss $\mathcal{L}$ is the weighted sum of $\mathcal{L}^{(\tau)}_{all}$ across all layers.

$$\mathcal{L} = \sum\nolimits_{\tau=1}^{\mathbf{N}_\tau} \varrho^{(\tau)}\mathcal{L}^{(\tau)}_{all} + \lambda\|\Theta\|_2^2 \tag{35}$$

$$\varrho^{(\tau)} = \begin{cases} 1/(\mathbf{N}_\tau+2) & \tau < \mathbf{N}_\tau \\ 3/(\mathbf{N}_\tau+2) & \tau = \mathbf{N}_\tau \end{cases} \tag{36}$$

where $\mathbf{N}_\tau$ represents the total number of layers in the MTML. λ represents the coefficient of $\mathbf{L}_2$-regularization, and Θ signifies the set of all trainable parameters.

4. Experiments

4.1. Datasets

We evaluate our MTML on the ECF dataset, which is derived from the MELD dataset [44] and augmented with cause annotations for emotions. The ECF dataset comprises 13,619 utterances and 9,794 annotated emotion-cause pairs across 1,374 conversations. Adopting the approach from [1], we partition the dataset into training, validation, and testing sets at the conversation level, using a ratio of 7:1:2.

4.2. Comparison models

We compare our MTML with the heuristic approaches and the two-step deep learning approach proposed by Wang et al. [1]. We also reproduce several representative deep learning methods in textual ECPE and adapt them to the multimodal scenario for comparison.

Heuristic Approach: These methods [1] offer varied approaches for identifying cause utterances, leveraging inherent patterns in the relative positions of causes and emotions.

Deep Learning Approach: Due to the novelty of the MC-ECPE task, there is little work in this field. These deep learning approaches are transferred from the textual ECPE domain. We concatenate the feature vectors of textual, visual, and audio modalities as sentence representations to enable the model to adapt to the multimodal scenario.

- MECPE-2steps [1]: The model first identifies emotions and causes, then constructs candidate pairs and filters them, completing the ECPE task in a pipeline paradigm.
- SLSN [7]: Extracts emotion-cause pairs directly by employing a local search mechanism within the sliding window, enabling simultaneous detection and matching of emotions and causes.
- A^2Net [39]: Addresses ECPE challenges by performing feature-task alignment to model specific and interactive features, and inter-task alignment to narrow the label distance between ECPE and the combinations of emotion extraction and cause extraction for better label consistency.
- SHARK [42]: Proposes a commonsense knowledge-enhanced generative framework that formulates the ECPE task as an index generation problem, enabling end-to-end generation of emotion-cause pairs using a sequence-to-sequence model.

Large Language Model (LLM): We conducted evaluations on the MC-ECPE task utilizing GPT-4 [49]. The corresponding prompt generally consists of speaker information + complete conversation (text + three evenly sampled frames from the video over time) + task descriptions for the three subtasks (emotion extraction, cause extraction, emotion-cause pairing) in MC-ECPE, as illustrated in Fig. 3, where $<$UTT_$i>$ denotes the text of the ith utterance, and $<$UTT_i_IMG_$j>$ represents the jth frame image corresponding to the ith utterance.

In our ablation study, we present variations of our proposed MTML: "**w/o** Multimodal information" indicates that the model uses only text as input. "**w/o** Audio" and "**w/o** Video" indicate that audio and video information are not used, respectively. "**w/o** MIGAT" denotes the model without MIGAT, using concatenation instead. "**w/o** Intra-modal connections", "**w/o** Cross-modal connections", and "**w/o** Cross-task connections" signify the absence of those edges. "**w/o** Cross-task connections($e \rightarrow c$)" and "**w/o** Cross-task connections($c \rightarrow e$)" indicate the removal of edges from the emotion extraction to the cause extraction or vice versa. "**w/o** $\mathbf{Expert}_s$" means excluding shared Experts, which implies that two tasks share information at every layer in a hard-sharing manner. "**w/o** $\widehat{\mathbf{Expert}}_c$", "**w/o** $\widehat{\mathbf{Expert}}_e$", and "**w/o** $\widehat{\mathbf{Expert}}_s$" signifies that the second-layer and subsequent Experts are identical to the first-layer ones. "**w/o** PMTL" indicates the removal of all experts and gating networks, meaning the entire PMTL module is eliminated. In this case, the emotion and cause vectors output by MIGAT are used separately for emotion extraction and cause extraction, and their concatenation is used for pair filtering.

Please play the role of a video expression classification expert. We will provide N utterances, each utterance containing speaker's content and three temporally uniformly sampled frames. All these utterances belong to the same conversation context. Please read these utterances first, and then you'll receive two tasks based on these utterances.

the content of utterance_1 is '<UTT_1>'
the frames of utterance_1 are shown below
<UTT_1_IMG_1>
<UTT_1_IMG_2>
<UTT_1_IMG_3>

the content of utterance_2 is '<UTT_2>'
the frames of utterance_2 are shown below
<UTT_2_IMG_1>
<UTT_2_IMG_2>
<UTT_2_IMG_3>

…

the content of utterance_N is '<UTT_N>'
the frames of utterance_N are shown below
<UTT_N_IMG_1>
<UTT_N_IMG_2>
<UTT_N_IMG_3>

Above are all the utterances. Please finish the first task based on them: For each utterance, please predict if the emotion of the speaker in the utterance is neutral emotion (if the emotion is not anger or disgust or fear or joy or sadness or surprise, then it is neutral emotion), if it is, please answer "yes". Otherwise, answer "no". There may be more than one person in each video frame, you only need to focus on the speaker in this video frame, not the other people in this video frame. Here is an example of your answer format: if you predict that utterance_A has non-neutral emotion, utterance_B has neutral emotion, you should output "utterance_A : no \n utterance_B : yes", in which "A" and "B" refer to the index of utterance.
After the first task is finished, please finish the second task: For each utterance, if your answer is "no" in the first task, please predict which utterance causes this emotion, if you cannot find a cause utterance, then randomly select an utterance as the cause. There may be more than one person in each video frame, you only need to focus on the speaker in this video frame, not the other people in this video frame. Here is an example of your answer format: if your answer for utterance_A in the first task is "no", and you think the emotion is caused by utterance_B, then your answer should be (utterance_A, utterance_B), in which "A" and "B" refer to the index of utterance, "A" is allowed to be smaller than "B". For the above two tasks, you only need to show your predictions, please don't show your analysis.

Fig. 3. Prompt template used to perform MC-ECPE.

4.3. Experimental settings

Hyperparameters in MTML are manually tuned for each dataset using hold-out validation, as shown in Table 1. The reported results in this paper are the average score obtained from 5 random runs on the test set. All experiments are conducted using a single NVIDIA RTX A6000 GPU.

Table 1
Detailed hyperparameter settings.

Hyperparameters	ECF
Batch size	8
Epochs	15
d_a	6373
d_v	4096
d_t	768
d_h	200
d_p	50
λ	0.00001
$\mathbf{N}_\tau$	3
$\mathbf{N}_\xi$	1
ω	1
The learning rate of BERT parameters	0.00001
The learning rate of other parameters	0.001
Dropout rate	0.2
Optimizer	Adam
Maximum conversation length	35
Maximum utterance length	35

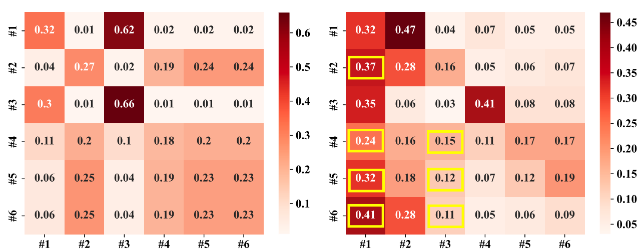

Fig. 4. Visualization of attention matrices for the emotion extraction in a conversation between Baseline (MECPE-2steps ($Text_{BERT}$) (left) and the MTML (right). Yellow boxes highlight corresponding causal utterances.

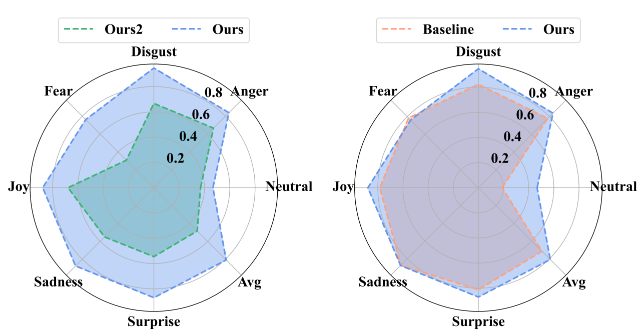

Fig. 5. F1 score for the cause extraction when emotion extraction is correct, with individual category F1 scores visualized and an overall F1 score represented as ‘Avg’. ‘Ours’ denotes the MTML. ‘Ours2’ signifies the performance of the MTML when emotion extraction is incorrect. We divide the utterances based on their emotion categories and evaluate the performance for each category to obtain the F1 score for each one.

4.4. Analysis of motivation

Cause extraction enhances emotion extraction: Our analysis indicates that prioritizing causal expressions improves emotion extraction accuracy. Among correctly classified utterances in the Baseline, causal utterances account for 19.77 % of the top-k attention scores (For the MTML, it’s $\mathcal{A}^\tau$, and for the Baseline, it’s the self-attention weight matrix in the Transformer [47].), compared to 14.18 % in incorrectly classified ones. The k denotes the number of causal utterances. By integrating explicit and implicit modeling strategies in the MTML, the proportion of causal utterances significantly rises to 36.82 % for correctly classified utterances and 15.92 % for misclassified ones. Notably, causal utterances are ranked on average at 3.5959 using attention scores in the MTML, compared to 4.7459 in the Baseline. Conversely, causal utterances are ranked on average at 4.9045 and 5.1927 in misclassified instances, respectively. These findings affirm our approach’s inclination towards prioritizing causal utterances during emotion extraction, as supported by attention score visualizations in Fig. 4. This enhances the model’s ability in the emotion extraction task.

Emotion extraction enhances cause extraction: Furthermore, emotions significantly influence the cause extraction, as evidenced by prior research [16]. Understanding the emotion behind an utterance greatly facilitates cause extraction. In Fig. 5, we compared cause extraction performance between correct and incorrect emotion extraction. Significantly lower cause extraction performance was observed when emotions were extracted incorrectly, reinforcing the crucial role of emotions in cause extraction. After modeling this dependency, we observed significant improvements over the Baseline across all scenarios except for ‘Fear’ and ‘Sadness’, as demonstrated in Fig. 5. This underscores the effectiveness of our MTML in leveraging emotions to improve cause extraction.

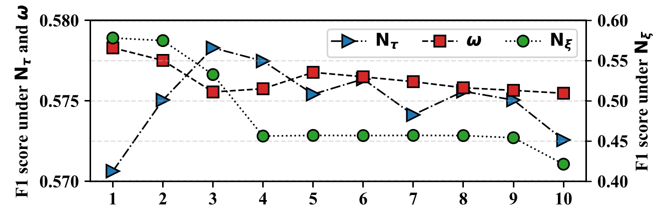

Fig. 6. The F1 score of the pair extraction on the validation set across various hyperparameter settings.

4.5. Overall results

In Table 2, we present our experimental results for the MTML, GPT4, four comparison methods, and a human performance test (following [1]) on the ECF dataset.

Our proposed MTML model achieves the best performance across all three tasks: emotion extraction, cause extraction, and pair extraction. Particularly, in the cause extraction and pair extraction tasks, our model outperforms existing methods with F1 score improvements of 3.02 % and 2.52 %, respectively, approaching human-level performance. This demonstrates the effectiveness and advancement of our method. In addition, GPT-4 shows significantly lower performance in the emotion-cause pair extraction task, likely due to its limitations in understanding and processing emotional semantics [50].

4.6. Hyper-parameter analysis

In Fig. 6, we illustrate the model’s performance on the ECF validation set by adjusting hyperparameters, including $\mathbf{N}_\xi$, $\mathbf{N}_\tau$, and ω. For the layer count $\mathbf{N}_\tau$ of the MTML, the model initially shows improvement, followed by a decrease and eventual stabilization with minor fluctuations. Importantly, performance consistently surpasses the case where the parameter is set to 1, demonstrating the effectiveness of our MTML across different hyperparameter settings. The number of layers in MI-GAT $\mathbf{N}_\xi$ shows a decreasing trend, while the window size ω exhibits an initial fluctuation followed by a decrease. This could be attributed to the reduced semantic content provided by distant utterances [51], leading to more noise with larger context windows [52]. Moreover, an increased layer count in GAT may result in over-smoothing of nodes [53].

4.7. Ablation study

In this section, we analyze the impact of various components in the MTML. Table 3 demonstrates that all components of the MTML have significantly improved results, reflected in metrics such as P, R, and F1. This is further supported by the statistical analysis, where the p-value $\ll$ is 0.05 for the paired t-test.

Table 2
Comparison results on different methods. We present the performance metrics for three subtasks, including precision (P), recall (R), and F1 score (F1). A '-' indicates that the method incorporates truth emotion annotations. Baselines marked with '*' are our re-implementation results using their open source code. The highest scores are highlighted in bold.

Methods		Emotion Extraction P	R	F1	Cause Extraction P	R	F1	Pair Extraction P	R	F1
LLM	GPT-4	**0.7865**	0.4588	0.5795	0.5406	0.3509	0.4256	0.2604	0.1175	0.1619
Heuristic	$E_{Annotation} + C_{Bernoulli}$	–	–	–	0.642	0.5485	0.5915	0.4958	0.3307	0.3967
	$E_{Annotation} + C_{Multinomial}$	–	–	–	0.6417	0.5483	0.5913	0.4951	0.3303	0.3963
Deep Learning	MECPE-2steps	0.7745	0.8110	0.7916	0.6842	0.7243	0.7027	0.5747	0.4981	0.5320
	SLSN*	0.6563	**0.9087**	0.7620	0.5807	**0.8402**	0.6864	0.4974	0.5202	0.5080
	A^{2}Net*	0.7354	0.8298	0.7790	0.6257	0.7890	0.6966	**0.5871**	0.3811	0.4621
	SHARK*	0.7474	0.8018	0.7728	0.6113	0.8065	0.6944	0.4817	**0.5692**	0.5212
	MTML (Ours)	0.7769	0.8171	**0.7965**	**0.6898**	0.7825	**0.7329**	0.5853	0.5322	**0.5572**
Human	Human #1	0.7211	0.8525	0.7813	0.6262	0.8440	0.7190	0.5073	0.6344	0.5635
	Human #2	0.7248	0.8760	0.7933	0.6252	0.8636	0.7253	0.545	0.6047	0.5724

Table 3
Ablation results of the MTML. We have considered the impact of five key aspects (Multimodal information, MIGAT, Implicit modeling, Explicit modeling and PMTL) on model performance (highlighted in grey) and reported the influence of their respective components.

Methods		Emotion Extraction P	R	F1	Cause Extraction P	R	F1	Pair Extraction P	R	F1
MTML		0.7769	0.8171	0.7965	0.6898	0.7825	0.7329	0.5853	0.5322	0.5572
Multimodal information	**w/o** Multimodal information	0.7695	0.8220	0.7946	0.7029	0.7475	0.7243	0.5945	0.5116	0.5495
	w/o Audio	0.7783	0.8090	0.7930	0.6992	0.7582	0.7268	0.5771	0.5293	0.5504
	w/o Video	0.7746	0.8169	0.7949	0.6878	0.7818	0.7305	0.5848	0.5284	0.5545
MIGAT	**w/o** MIGAT	0.7752	0.8099	0.7918	0.6517	0.7640	0.7024	0.5796	0.5121	0.5421
	w/o Cross-modal connections	0.7706	0.8159	0.7923	0.6896	0.7786	0.7302	0.5830	0.5271	0.5530
	w/o Intra-modal connections	0.7765	0.8134	0.7943	0.6875	0.7763	0.7287	0.5786	0.5309	0.5526
	w/o Cross-task connections	0.7729	0.8144	0.7927	0.6766	0.7786	0.7234	0.5776	0.5291	0.5502
Implicit modeling	**w/o** Implicit modeling	0.7767	0.8076	0.7908	0.6874	0.7669	0.7245	0.5774	0.5149	0.5410
	w/o Cross-task connections$_{(e\rightarrow c)}$	0.7757	0.8170	0.7955	0.6886	0.7699	0.7263	0.5821	0.5259	0.5511
	w/o Cross-task connections$_{(c\rightarrow e)}$	0.7748	0.8162	0.7941	0.6882	0.7757	0.7282	0.5823	0.5273	0.5530
	w/o $\mathbf{Expert}_s$	0.7744	0.8138	0.7923	0.6826	0.7770	0.7262	0.5667	0.5300	0.5461
	w/o $\widehat{\mathbf{Expert}}_s$	0.7753	0.8129	0.7933	0.6822	0.7801	0.7273	0.5767	0.5293	0.5518
Explicit modeling	**w/o** Explicit modeling	0.7732	0.8119	0.7918	0.6890	0.7611	0.7226	0.5808	0.5141	0.5443
	w/o $\widehat{\mathbf{Expert}}_e$	0.7729	0.8129	0.7920	0.6874	0.7750	0.7274	0.5776	0.5296	0.5515
	w/o $\widehat{\mathbf{Expert}}_c$	0.7766	0.8130	0.7937	0.6870	0.7661	0.7242	0.5782	0.5287	0.5503
PMTL	**w/o** PMTL	0.7728	0.8105	0.7911	0.6851	0.7293	0.7063	0.5769	0.5079	0.5396

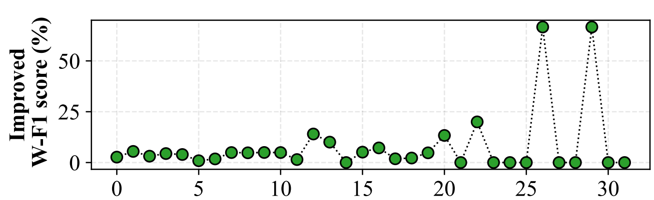

Fig. 7. Lifting F1 score of the pair extraction on different positions in conversations. The noticeable jump in F1 score at positions 26 and 29 (both have 2 pairs) stems from limited data amplifying performance improvements.

Analysis of Multimodal information: We eliminate audio, video, and both modalities from multimodal information, resulting in performance degradation. These experimental results highlight the value of multimodal information in improving the understanding of conversations and accurately identifying emotions and their causes.

Analysis of MIGAT: Existing efforts for the MC-ECPE task rely solely on traditional fusion methods such as concatenation, which limit context understanding between modalities. In response, we propose the MIGAT, incorporating three connections: intra-modal connections for modeling context within the same modality, cross-modal connections for learning multimodal complementarity, and cross-task connections for obtaining shared information from the correlated task. The results shown in Fig. 7 indicate improved performance over longer contexts, and the results in Table 3 demonstrate a significant improvement of 1.51 % in the F1 score for pair extraction with MIGAT, thereby validating the effectiveness of this module. Specifically, the first two connections enhance conversation understanding in multimodal environments, while the third belongs to the implicit modeling strategy. Intra-modal connections slightly outperform cross-modal connections in pair extraction, underscoring the importance of modeling context in the MC-ECPE task.

Analysis of Implicit modeling: In the MTML, the implicit modeling strategy leverages insights from shared Experts and cross-task connections MIGAT. Unlike some other multi-task ECPE methods [17], we gradually decouple shared and task-specific components at higher levels to prevent boosting one task at the expense of another. Results in Table 3 further support this approach, demonstrating simultaneous improvements across three tasks, with a 1.62 % increase in F1 score for the pair extraction task. Additionally, the emotion and cause extraction mutually benefit as they are inherently interdependent tasks. Therefore, we introduced cross-task connections and observed that enhancing cause extraction from emotion extraction yields greater improvements compared to the reverse. This may be attributed to the models' current strength in emotion extraction, leading to a more pronounced enhancement.

Analysis of Explicit modeling: Explicit modeling involves extracting emotional and causal probability distributions from lower layers to enhance reasoning at the current layer. Compared to obtaining shared information from implicit modeling strategies, this explicit distillation process exhibits a 1.29 % improvement in pair extraction, surpassing shared Experts in effectiveness. Interestingly, a similar trend is observed in explicit modeling compared to implicit modeling: utilizing the probability distribution from emotion extraction for cause extraction yields superior results. Furthermore, explicit modeling demonstrates a more

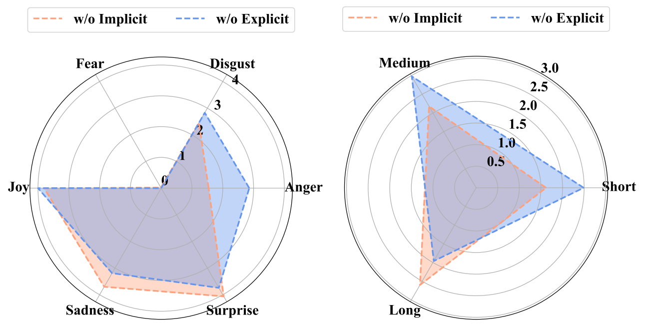

Fig. 8. Decreasing F1 score (in percentage points) for pair extraction. We divide ground truth pairs and model-predicted pairs based on the emotion category of the emotion utterance in each pair, and calculate the performance for each category.

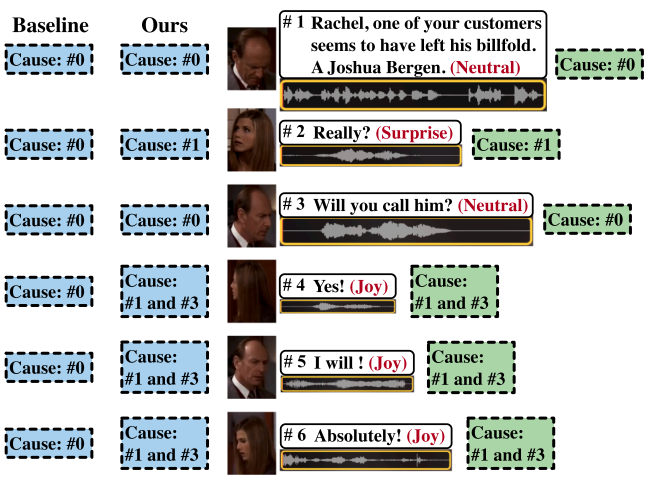

Fig. 9. Examples of utterances from the ECF dataset for the case study. Emotions corresponding to predicted causes are highlighted in blue dashed boxes, with #0 denoting neutral emotions without corresponding causes.

substantial enhancement in the cause extraction task compared to implicit modeling, achieving a 1.03 % increase in the F1 score.

Analysis of PMTL: PMTL incorporates task-specific and task-shared experts, leveraging gate networks to aggregate knowledge from different experts for various tasks. Additionally, PMTL enhances current-layer reasoning by utilizing predictions from the previous layer. As shown in Table 3, removing PMTL results in substantial performance degradation across all tasks, particularly in cause extraction and pair extraction, where F1 scores drop by 2.66 % and 1.76 %, respectively, demonstrating the critical role of PMTL.

4.8. Complementarity analysis

In this section, we explore the complementarity between implicit and explicit modeling strategies in the MTML framework. As shown in Fig. 8, implicit modeling excels in long conversations, particularly in eliciting 'Sadness' and 'Surprise' emotions, while explicit modeling demonstrates strong performance in short to medium conversations, notably in evoking 'Disgust' and 'Anger' emotions. Each module showcases distinct strengths, highlighting their complementary performance across different conversations, as further evidenced in Table 3.

4.9. Case study

In Fig. 9, we select a conversation from the test set of the ECF dataset as a case. In this conversation, Mr. Waltham informs Rachel that his customer, Joshua Bergen, seems to have forgotten his wallet. Rachel

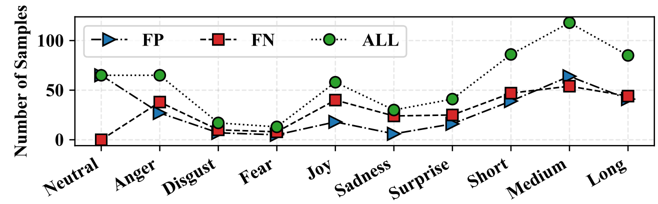

Fig. 10. Error analysis across various emotional categories and Conversation lengths (Short: first 1/3, Medium: middle 1/3, and Long: last 1/3 of a conversation). FP: Misclassify non-true pairs as positive, while MECPE-2steps ($Text_{BERT}$) gets them right as negative. FN: Misclassify true pairs as negative, while MECPE-2steps ($Text_{BERT}$) gets them right as positive. ALL: Sum of both cases.

expresses surprise upon hearing this, and Mr. Waltham asks if she can call Joshua. Rachel agrees and eagerly states that she will contact him immediately. Due to the straightforward and concise nature of the conversation, both the Baseline and our MTML achieved perfect results in emotion extraction. However, our method, benefiting from the implicit and explicit modeling of mutual dependencies between emotion and cause extraction, pays more attention to the cause utterances causing the emotion compared to the Baseline, as illustrated in Fig. 4. Furthermore, while the Baseline fails to identify any cause utterance in this conversation for cause extraction, our approach accurately distinguishes all cause utterances, thus demonstrating outstanding performance in the emotion-cause pair extraction task.

4.10. Error analysis

In Fig. 10, we analyze instances where the MTML model misclassifies samples that were correctly predicted by MECPE-2steps (TextBERT), totaling 289 samples in the ECF dataset. Notably, it struggles particularly in Medium positions of conversations (111 samples), with relatively better performance observed in Short (86 samples) and Long (85 samples) positions. This could be attributed to the advantages of MIGAT. Specifically, it shows improved performance in fewer categories such as 'Disgust' and 'Fear'. This might be because the MTML model better captures the correlation between emotions when handling these emotion categories, rather than being limited to individual emotion representations.

5. Conclusion

Emotion extraction and cause extraction are closely related tasks within MC-ECPE. Conversational context dependency, multi-modal interaction, and multi-task correlation are three pivotal factors that contribute to these tasks, thereby enhancing emotion-cause pair extraction. In this paper, we propose the MTML framework to comprehensively model the mutual dependency between emotion and cause extraction tasks. Specifically, we employ both implicit and explicit modeling strategies: (1) Implicit modeling utilizes shared Experts and cross-task connections within a MIGAT to extract shared information from the correlated task. MIGAT is designed to capture intra-modal and cross-modal dependencies, as well as cross-task dependencies. (2) Explicit modeling involves a progressive separation routine to extract emotional and causal logits from the lower layer, serving as a form of knowledge to enhance reasoning at the current layer. Experimental results on the ECF dataset demonstrate the effectiveness of both implicit and explicit modeling, as well as their complementary nature. Furthermore, our MTML has achieved a new state-of-the-art performance.

CRediT authorship contribution statement

Geng Tu: Validation, Investigation, Conceptualization, Writing - original draft, Data curation, Methodology, Writing - review & editing; **Jun Wang:** Writing - original draft, Validation, Conceptualization, Data curation, Methodology; **Li Yang:** Resources, Investigation, Visualization; **Bin Liang:** Supervision, Writing - review & editing, Conceptualization; **Erik Cambria:** Supervision, Writing - review & editing; **Wenjie Li:** Supervision, Writing - review & editing, Funding acquisition; **Ruifeng Xu:** Funding acquisition, Writing - review & editing, Software, Resources, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and there is no professional or other personal interest of any nature or kind in any product, service, and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled "Multi-Task Mutual Learning for Multimodal Emotion-Cause Pair Extraction in Conversations".

Acknowledgments

We thank all anonymous reviewers for their helpful comments. This work was supported in part by the National Natural Science Foundation of China under Grants 62176076, theNatural Science Foundation of Guangdong under Grant 2023A1515012922, Shenzhen Foundational Research Funding under Grant JCYJ20220818102415032, The Major Key Project of PCL under Project PCL2023A09, Guangdong Provincial Key Laboratory of Novel Security Intelligence Technologies under Grant 2022B1212010005k, and Research Grants Council of Hong Kong under Grant 15213323.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.inffus.2025.103877

References

[1] F. Wang, Z. Ding, R. Xia, Z. Li, J. Yu, Multimodal emotion-cause pair extraction in conversations, IEEE Trans. Affect. Comput. 14 (03) (2023) 1832–1844.
[2] R. Xia, Z. Ding, Emotion-cause pair extraction: a new task to emotion analysis in texts, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, 2019, pp. 1003–1012.
[3] L. Gui, R. Xu, D. Wu, Q. Lu, Y. Zhou, Event-driven emotion cause extraction with corpus construction, in: Social Media Content Analysis: Natural Language Processing and Beyond, World Scientific, 2018, pp. 145–160.
[4] W. Li, Y. Li, V. Pandelea, M. Ge, L. Zhu, E. Cambria, ECPEC: emotion-cause pair extraction in conversations, IEEE Trans. Affect. Comput. 14 (3) (2022) 1754–1765.
[5] Y. Chen, W. Hou, S. Li, C. Wu, X. Zhang, End-to-end emotion-cause pair extraction with graph convolutional network, in: Proceedings of the 28th International Conference on Computational Linguistics, 2020, pp. 198–207.
[6] Z. Cheng, Z. Jiang, Y. Yin, C. Wang, S. Ge, Q. Gu, A consistent dual-mrc framework for emotion-cause pair extraction, ACM Trans. Inf. Syst. 41 (4) (2023) 1–27.
[7] Z. Cheng, Z. Jiang, Y. Yin, H. Yu, Q. Gu, A symmetric local search network for emotion-cause pair extraction, in: Proceedings of the 28th International Conference on Computational Linguistics, 2020, pp. 139–149.
[8] Z. Ding, R. Xia, J. Yu, End-to-end emotion-cause pair extraction based on sliding window multi-label learning, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, 2020, pp. 3574–3583.
[9] Z. Ding, R. Xia, J. Yu, ECPE-2D: emotion-cause pair extraction based on joint two-dimensional representation, interaction and prediction, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 2020, pp. 3161–3170.
[10] C. Fan, C. Yuan, J. Du, L. Gui, M. Yang, R. Xu, Transition-based directed graph construction for emotion-cause pair extraction, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 2020, pp. 3707–3717.
[11] P. Wei, J. Zhao, W. Mao, Effective inter-clause modeling for end-to-end emotion-cause pair extraction, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 2020, pp. 3171–3181.
[12] C. Yuan, C. Fan, J. Bao, R. Xu, Emotion-cause pair extraction as sequence labeling based on a novel tagging scheme, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, 2020, pp. 3568–3573.
[13] C. Fan, C. Yuan, L. Gui, Y. Zhang, R. Xu, Multi-task sequence tagging for emotion-cause pair extraction via tag distribution refinement, IEEE/ACM Trans. Audio Speech Lang. Process. 29 (2021) 2339–2350.
[14] X. Chen, Q. Li, J. Wang, A unified sequence labeling model for emotion cause pair extraction, in: Proceedings of the 28th International Conference on Computational Linguistics, 2020, pp. 208–218.
[15] Z. Cheng, Z. Jiang, Y. Yin, N. Li, Q. Gu, A unified target-oriented sequence-to-sequence model for emotion-cause pair extraction, IEEE/ACM Trans. Audio Speech Lang. Process. 29 (2021) 2779–2791.
[16] A. Singh, S. Hingane, S. Wani, A. Modi, An end-to-end network for emotion-cause pair extraction, in: Proceedings of the Eleventh Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis, 2021, pp. 84–91.
[17] S. Wu, F. Chen, F. Wu, Y. Huang, X. Li, A multi-task learning neural network for emotion-cause pair extraction, in: Proceedings of the 24th European Conference on Artificial Intelligence, 2020, pp. 2212–2219.
[18] Z. Wu, X. Dai, R. Xia, Pairwise tagging framework for end-to-end emotion-cause pair extraction, Front. Comput. Sci. 17 (2) (2023) 172314.
[19] P. Zhu, B. Wang, K. Tang, H. Zhang, X. Cui, Z. Wang, A knowledge-guided graph attention network for emotion-cause pair extraction, Knowl. Based Syst. 286 (2024) 111342.
[20] Y. Wei, X. Wang, L. Nie, X. He, R. Hong, T.-S. Chua, MMGCN: multi-modal graph convolution network for personalized recommendation of micro-video, in: Proceedings of the 27th ACM International Conference on Multimedia, 2019, pp. 1437–1445.
[21] F. Chen, J. Shao, S. Zhu, H.T. Shen, Multivariate, multi-frequency and multimodal: rethinking graph neural networks for emotion recognition in conversation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 10761–10770.
[22] S.Y.M. Lee, Y. Chen, C.-R. Huang, A text-driven rule-based system for emotion cause detection, in: Proceedings of the North American Chapter of the Association for Computational Linguistics 2010 Workshop on Computational Approaches to Analysis and Generation of Emotion in Text, 2010, pp. 45–53.
[23] D. Ghazi, D. Inkpen, S. Szpakowicz, Detecting emotion stimuli in emotion-bearing sentences, in: Proceedings of the 16th International Conference on Computational Linguistics and Intelligent Text Processing, 2015, pp. 152–165.
[24] S. Song, Y. Meng, Detecting concept-level emotion cause in microblogging, in: Proceedings of the 24th International Conference on World Wide Web, 2015, pp. 119–120.
[25] A. Neviarouskaya, M. Aono, Extracting causes of emotions from text, in: Proceedings of the 6th International Joint Conference on Natural Language Processing, 2013, pp. 932–936.
[26] W. Li, H. Xu, Text-based emotion classification using emotion cause extraction, Expert Syst. Appl. 41 (4) (2014) 1742–1749.
[27] K. Gao, H. Xu, J. Wang, Emotion cause detection for chinese micro-blogs based on ecocc model, in: Advances in Knowledge Discovery and Data Mining, 2015, pp. 3–14.
[28] S. Yada, K. Ikeda, K. Hoashi, K. Kageura, A bootstrap method for automatic rule acquisition on emotion cause extraction, in: Proceedings of the 2017 IEEE International Conference on Data Mining Workshops, 2017, pp. 414–421.
[29] Y. Chen, S.Y.M. Lee, S. Li, C.-R. Huang, Emotion cause detection with linguistic constructions, in: Proceedings of the 23rd International Conference on Computational Linguistics, 2010, pp. 179–187.
[30] I. Russo, T. Caselli, F. Rubino, E. Boldrini, P. Martínez-Barco, et al., Emocause: an easy-adaptable approach to emotion cause contexts, in: Proceedings of the 2nd Workshop on Computational Approaches to Subjectivity and Sentiment Analysis, 2011, pp. 153–160.
[31] L. Gui, L. Yuan, R. Xu, B. Liu, Q. Lu, Y. Zhou, Emotion cause detection with linguistic construction in chinese weibo text, in: Proceedings of the 3rd CCF International Conference on Natural Language Processing and Chinese Computing, 2014, pp. 457–464.
[32] R. Xu, J. Hu, Q. Lu, D. Wu, L. Gui, An ensemble approach for emotion cause detection with event extraction and multi-kernel SVMS, Tsinghua Sci. Technol. 22 (6) (2017) 646–659.
[33] L. Gui, J. Hu, Y. He, R. Xu, Q. Lu, J. Du, A question answering approach for emotion cause extraction, in: Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing, 2017, pp. 1593–1602.
[34] X. Li, K. Song, S. Feng, D. Wang, Y. Zhang, A co-attention neural network model for emotion cause analysis with emotional context awareness, in: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, 2018, pp. 4752–4757.
[35] X. Yu, W. Rong, Z. Zhang, Y. Ouyang, Z. Xiong, Multiple level hierarchical network-based clause selection for emotion cause extraction, IEEE Access 7 (2019) 9071–9079.
[36] B. Xu, H. Lin, Y. Lin, Y. Diao, L. Yang, K. Xu, Extracting emotion causes using learning to rank methods from an information retrieval perspective, IEEE Access 7 (2019) 15573–15583.
[37] Z. Ding, H. He, M. Zhang, R. Xia, From independent prediction to reordered prediction: integrating relative position and global label information to emotion cause identification, in: Proceedings of the AAAI Conference on Artificial Intelligence, 33, 2019, pp. 6343–6350.
[38] H. Feng, J. Liu, J. Zheng, H. Chen, X. Shang, Q. Ma, Joint constrained learning with boundary-adjusting for emotion-cause pair extraction, in: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics, 2023, pp. 1118–1131.
[39] S. Chen, X. Shi, J. Li, S. Wu, H. Fei, F. Li, D. Ji, Joint alignment of multi-task feature and label spaces for emotion cause pair extraction, in: Proceedings of the 29th International Conference on Computational Linguistics, 2022, pp. 6955–6965.

[40] G. Hu, Y. Zhao, G. Lu, Emotion prediction oriented method with multiple supervisions for emotion-cause pair extraction, IEEE/ACM Trans. Audio Speech Lang. Process. 31 (2023) 1141–1152.

[41] X. Song, D. Zou, Y. Yu, X. Zhang, FW-ECPE: an emotion-cause pair extraction model based on fusion word vectors, in: Proceedings of the 2023 International Joint Conference on Neural Networks, 2023, pp. 1–7.

[42] F. Wang, J. Yu, R. Xia, Generative emotion cause triplet extraction in conversations with commonsense knowledge, in: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 2023, pp. 3952–3963.

[43] J.D. M.-W.C. Kenton, L.K. Toutanova, BERT: pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics, 2019, pp. 4171–4186.

[44] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, R. Mihalcea, MELD: a multimodal multi-party dataset for emotion recognition in conversations, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, 2019, pp. 527–536.

[45] F. Eyben, M. Wöllmer, B. Schuller, Opensmile: the munich versatile and fast open-source audio feature extractor, in: Proceedings of the 18th ACM International Conference on Multimedia, 2010, pp. 1459–1462.

[46] S. Ji, W. Xu, M. Yang, K. Yu, 3D convolutional neural networks for human action recognition, IEEE Trans. Pattern Anal. Mach. Intell. 35 (1) (2012) 221–231.

[47] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, in: Advances in Neural Information Processing Systems, 30, 2017.

[48] J. Su, M. Ahmed, Y. Lu, S. Pan, W. Bo, Y. Liu, Roformer: enhanced transformer with rotary position embedding, Neurocomputing 568 (2024) 127063.

[49] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F.L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al., Gpt-4 technical report, arXiv preprint arXiv:2303.08774 (2023).

[50] W. Zhao, Y. Zhao, X. Lu, S. Wang, Y. Tong, B. Qin, Is chatgpt equipped with emotional dialogue capabilities?, arXiv preprint arXiv:2304.09582 (2023).

[51] P. Zhong, D. Wang, C. Miao, Knowledge-enriched transformer for emotion detection in textual conversations, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, 2019, pp. 165–176.

[52] G. Tu, B. Liang, R. Mao, M. Yang, R. Xu, Context or knowledge is not always necessary: a contrastive learning framework for emotion recognition in conversations, in: Findings of the Association for Computational Linguistics: ACL 2023, 2023, pp. 14054–14067.

[53] H. Tang, J. Liu, S. Yan, R. Yan, Z. Li, J. Tang, M3Net: multi-view encoding, matching, and fusion for few-shot fine-grained action recognition, in: Proceedings of the 31st ACM International Conference on Multimedia, 2023, pp. 1719–1728.